

# Nanobody Discovery Summer School: Current & Future Methods



 23–25 Sep 2026 ·  University of Nova Gorica



|                          |                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Day 1 · Wed 23/09</b> | <b>EPICenter</b> , Nova Gorica (next to the railway station): <b>Kolodvorska pot 10, Nova Gorica</b> . Registration opens <b>09:00 with coffee and some snacks</b> . |
| <b>Day 2–3 · Thu–Fri</b> | <b>University of Nova Gorica, Rožna Dolina campus</b> , Vipavska 13, Rožna Dolina, 5000 Nova Gorica.                                                                 |
| <b>City buses</b>        | Lines <b>1, 2, 3 and 4 are free for everyone</b> . No ticket, no card, just get on.                                                                                  |
| <b>Wi-Fi</b>             | Network details will be given on site.                                                                                                                               |
| <b>Contacts</b>          | Dr. Marco Orlando: marco.orlando@ung.si · +39 338 687 4438   Dr. Klara Kropivšek Brumat: +386 41 908 698                                                             |
| <b>Emergency</b>         | <b>112</b> (ambulance, fire, police)                                                                                                                                 |

## Your accommodation

|                            |                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Solkan</b>              | <b>Garage Apartments</b> , Prvomajska 118C, 5250 Solkan · Apartma studio A · door code sent privately |
| <b>Šempeter pri Gorici</b> | <b>AirBnB</b> , Cesta Goriške fronte 84c, Šempeter pri Gorici, Nova Gorica 5290, Slovenia             |

Accommodation is booked for 22–26 September.

## Day 1 (Wed 23/09): getting to EPICenter

**Important:** Google Maps does not yet know about the **new pedestrian underpass under the railway**. It will send you on a long detour. Follow the orange route on the map below instead.

### From Šempeter pri Gorici

1. Take the city bus from **Šempeter Trg** in the direction of **Solkan**. Recommended: **08:40** (latest; arrives ~08:55).
2. Ride ~15 min (9 stops) and get off at **Nova Gorica Elektro G** (on Erjavčeva ulica).
3. Walk ~5 min, as described below.

### From Solkan (Garage Apartments, Prvomajska 118C)

1. Easiest is to walk – 16 min (GoogleMaps knows this path). We recommend walking or getting a public e-bike, since the bus will take much longer on this route.
2. Line 1 runs along **Prvomajska ulica**. From the stop **Solkan Sabotin** (opposite Hotel Sabotin), take the bus toward **Nova Gorica / Šempeter**. Recommended: 8:00 or 9:00.
3. Get off at **Prvomajska** at the Prvomajska roundabout.
4. Walk the last part as described below.

### E-bikes

You can get information on bike rental and exploring the areas around both Nova Gorica and Italian Gorizia here: <https://www.nomago.si/en/bikes/gorica-go2go>

A cycling path connects Šempeter pri Gorici to Solkan, and then continues along the Soča River (all the way to Plave). You can ask for more information anytime.

## Day 1: walking route bus stop → EPICenter (~5 min)

Google Maps will **not** show this route yet: the underpass under the railway is new, as the railway station was renovated last year. The Google Maps image is not yet up to date.

1. From **Elektro G**, walk **north along Prvomajska ulica** towards the roundabout.
2. Just before the roundabout, **turn left** onto the path across the open field towards the railway.
3. Go through the **new underpass** under the tracks.
4. On the other side, you are on **Kolodvorska pot**, by the railway station. **Turn right** and walk north past the old steam locomotive to **EPICenter**.



Orange line = walking route from the Elektro G bus stop to EPICenter (circled).

## Days 2–3 (Thu & Fri): getting to the UNG campus

The Rožna Dolina campus is on the bus route between Šempeter and Nova Gorica (lines 1, 2 and 4). Get off at **Meja or Politehnika** (if coming from the direction of Solkan).

- **From Šempeter:** bus from Šempeter Trg towards Solkan at **08:05** or **08:40**, a few minutes' ride. Or walk (~30 min).
- **From Solkan:** bus towards Šempeter at **08:00** from Solkan obračališče (the bus passes the **Nova Gorica Gostol** stop a few minutes later, cca 8.06), ~15 min ride.

### Thursday dinner

- **18:00:** we all go together by **minibus from the Rožna Dolina campus to Hiša Marica, Šmartno** (Goriška Brda).
- Return: minibus back at around 9 PM to different drop-off points: Solkan, Šempeter pri Gorici,...

## City bus timetable (Nomago, valid 1 Sep – 30 Jun)

Free on lines 1–4. Times are **departures from the first stop** of each direction; stops further along the route are passed a few minutes later. Line 1 runs via Prvomajska ulica (Elektro G); line 2 via Cankarjeva ulica (see map attached below). Check the line number on the bus.

### Monday – Friday

| From                            | Departures                                                                                                                                                                                                                 |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Šempeter Trg → Solkan           | 6:05, 6:40, 7:10, 7:20, <b>8:05, 8:40</b> , 9:05, 9:40, 10:05, 10:40, 11:10, 11:30, 12:05, 12:45, 13:05, 13:35, 14:20, 14:40, 15:05, 15:25, 16:05, 17:05, 17:20, 18:05, 19:05, 19:20, 20:05, 21:05                         |
| Solkan obračališče → Šempeter   | 5:30, 6:00, 6:05, 6:35, 7:00, 7:30, <b>8:00, 8:30</b> , 9:00, 9:30, 10:00, 10:30, 11:00, 11:30, 12:00, 12:40, 13:10, 13:30, 14:00, 14:30, 14:40, 15:05, 15:40, 16:30, 17:00, 17:30, 18:30, 19:00, 19:30, 20:30             |
| Nova Gorica (centre) → Šempeter | 5:39, 6:09, 6:14, 6:35, 6:44, 7:09, 7:39, 8:09, 8:39, 9:09, 9:39, 10:09, 10:39, 11:09, 11:39, 12:09, 12:49, 13:19, 13:39, 14:09, 14:10, 14:39, 14:49, 15:14, 15:22, 15:49, 16:39, 17:09, 17:39, 18:39, 19:09, 19:39, 20:39 |
| Nova Gorica (centre) → Solkan   | 6:17, 6:52, 7:22, 7:32, 7:33, 8:17, 8:52, 9:17, 9:52, 10:17, 10:52, 11:22, 11:42, 12:17, 12:57, 13:17, 13:47, 14:32, 14:52, 15:17, 15:37, 16:17, 17:17, 17:32, 18:17, 19:17, 19:32, 20:17, 21:17                           |

### Saturday, Sunday & holidays (hourly)

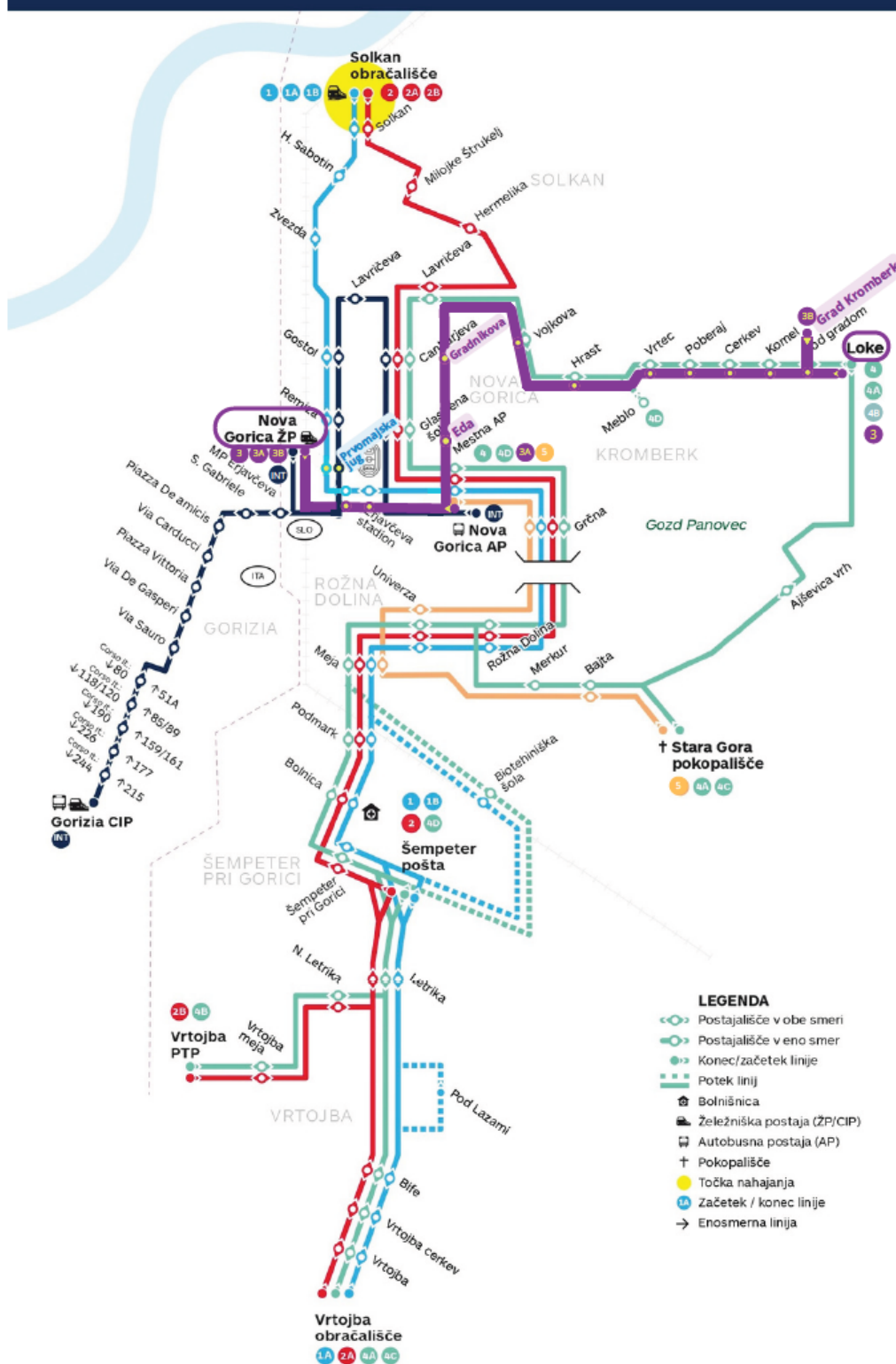
| From                            | Departures                            |
|---------------------------------|---------------------------------------|
| Šempeter Trg → Solkan           | every hour at :05, from 7:05 to 21:05 |
| Solkan obračališče → Šempeter   | every hour at :30, from 6:30 to 20:30 |
| Nova Gorica (centre) → Šempeter | every hour at :39, from 6:39 to 20:39 |
| Nova Gorica (centre) → Solkan   | every hour at :17, from 7:17 to 21:17 |

Source: [nomago.si](http://nomago.si), Mestni promet Nova Gorica. Live timetables: [nomago.si](http://nomago.si) → Vozni red.



# Mestni avtobus

## Shema linij



### LINIJ

- 1 SOLKAN OBRAČALIŠČE - ŠEMPETER POŠTA
- 1A SOLKAN OBRAČALIŠČE - VRTOJBA OBRAČALIŠČE
- 2 ŠEMPETER POŠTA - SOLKAN OBRAČALIŠČE (via Biotehniška)
- 2A SOLKAN OBRAČALIŠČE - ŠEMPETER POŠTA
- 2B SOLKAN OBRAČALIŠČE - VRTOJBA OBRAČALIŠČE

### LEGENDA

- Postajališče v obe smeri
- Postajališče v eno smer
- Konec/začetek linije
- Potek linij
- Bolnišnica
- Železniška postaja (ŽP/CIP)
- Autobusna postaja (AP)
- Pokopališče
- Točka nahajanja
- Začetek / konec linije
- Enosmerna linija

- 4 MESTNA AP - LOKE
- 4A LOKE / STARA GORA - VRTOJBA OBRAČALIŠČE (via Meblo)
- 4B LOKE - VRTOJBA PTP
- 4C VRTOJBA OBRAČALIŠČE - STARA GORA POKOPALIŠČE (via Biotehniška, Nova Gorica, Loke)

## Programme

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### Wednesday 23/09 · EPICenter

| Time          | Session                                                                             |
|---------------|-------------------------------------------------------------------------------------|
| 09:00         | Registration & coffee                                                               |
| 10:00         | Welcome & introduction to nanobody technology · Prof. Ario de Marco                 |
| 11:00         | Nanobody in silico screening from pre-immune libraries · Dr. Klara Kropivšek Brumat |
| 12:00         | Lunch break                                                                         |
| 13:30         | Nanobody de novo design with AI · Dr. Marco Orlando                                 |
| 15:15         | Coffee break (beverages only)                                                       |
| 15:30         | Nanobody panning against a target antigen · Dr. Claudia d'Ercole                    |
| 16:00 – 18.00 | Apero: networking & social                                                          |

### Thursday 24/09 · UNG Rožna Dolina campus

During the lab sessions, participants are split into three groups, supervised by **Mirna Nakić**, **Lucia Cikatricisová** and **Claudia d'Ercole**:

**Mirna Nakić's group**: Bor Krajnik, Nikol Mraz, Chaeyeon Park

**Lucia Cikatricisová's group**: Mariagrazia Scarano, Taehee Yu, Vicky Chouhan

**Claudia d'Ercole's group**: Gašper Vrečer, Perveen Akhtar, Boubek Saifi

| Time  | Session                                                                                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09:00 | Practical in vitro nanobody panning (lab)                                                                                                                                                  |
| 10:30 | Coffee break                                                                                                                                                                               |
| 11:30 | De novo designed nanobodies for intracellular protein perturbation · Bor Krajnik (Participant Talks — Session I)                                                                           |
| 11:50 | Generation and Characterization of Disease-Targeting Nanobodies · Boubek Saifi (Participant Talks — Session I)                                                                             |
| 12:10 | Engineered DARPIn-Based Imaging Probe Platform for the Detection of Pathological Tau and Tau Variants · Chaeyeon Park (Participant Talks — Session I)                                      |
| 12:30 | Lunch break                                                                                                                                                                                |
| 13:30 | Practical in vitro nanobody panning (lab)                                                                                                                                                  |
| 15:00 | Coffee break                                                                                                                                                                               |
| 15:30 | Characterization of Protein–Secondary Metabolite Interactions and Identification of Molecular Binding Determinants · Gašper Vrečer (Participant Talks — Session II)                        |
| 15:50 | Functional and Structural Characterisation of Anti-Fyn SH3 VHH and CDR3 Optimization for Improved Intracellular Binding · Mariagrazia Scarano (Participant Talks — Session II)             |
| 16:10 | Development of a Recombinant Nanobody Pipeline Integrating Computational Design and Phage Display for Sensitive Detection of Microcystin · Perveen Akhtar (Participant Talks — Session II) |

|              |                                                                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>16:30</b> | Discovery of TfR1-Binding Fab Fragments · Taehee Yu (Participant Talks — Session II)                                                            |
| <b>16:50</b> | Plant Epigenetics in Conifers: Identification of Epigenetic Biomarkers Under Stress Conditions · Vicky Chouhan (Participant Talks — Session II) |
| <b>17:10</b> | Free time: explore around                                                                                                                       |
| <b>17:30</b> | Conference dinner at Hiša Marica, Šmartno: minibus departs from the University – Rožna Dolina campus                                            |

### Friday 25/09 · UNG Rožna Dolina campus

| <b>Time</b>  | <b>Session</b>                                                                                         |
|--------------|--------------------------------------------------------------------------------------------------------|
| <b>09:00</b> | Molecular dynamics-based evolution of epitope-specific nanobodies · Asst. Prof. Sara Fortuna           |
| <b>10:00</b> | Coffee break                                                                                           |
| <b>10:30</b> | Data quality challenges and structural artefacts in nanobody–antigen databases · Asst. Prof. San Hadži |
| <b>11:30</b> | Final regards and discussion                                                                           |
| <b>12:00</b> | Lunch break                                                                                            |
| <b>13:00</b> | End of the Summer School                                                                               |

## Abstracts

Wednesday 23/09 · EPICenter

### Prof. Ario de Marco — Welcome & Introduction on Nanobody Technology

Wed 23/09, 10:00 · University of Nova Gorica · Opening Lecture

This introductory session provides a foundational overview of nanobody (VHH) libraries and their role in binder discovery. Participants will learn the key differences between immune libraries (from antigen-immunized animals), naïve or pre-immune libraries (from non-immunized sources), and synthetic libraries (constructed through targeted sequence design and CDR randomization). The session explores how library diversity is generated—through methods such as PCR amplification of VHH genes or synthetic variation—and how it influences library quality and downstream success. Common library sizes are discussed, as well as how different display platforms (e.g. phage, yeast, ribosome) are used to present libraries to antigens, with an emphasis on phage display.

### Dr. Klara Kropivsek — Nanobody in silico Screening from Pre-Immune Libraries

Wed 23/09, 11:00 · University of Nova Gorica · Lecture & Hands-on

Recent advances in high-throughput sequencing and structure prediction have enabled “virtual panning” of naïve nanobody (VHH) repertoires. This session reviews the core modules of computational nanobody discovery: repertoire profiling via MiXCR and clonotype clustering, homology modeling with NanobodyBuilder2 and ColabFold/Boltz-2, and docking benchmarking of published VHH–antigen complexes using ZDock and Boltz-2, scored by different scoring strategies. We'll showcase case studies from ongoing work—including clonotype analysis of a naïve llama library and redocking of SAbDab-derived VHH–antigen pairs to assess predictive performance.

### Dr. Marco Orlando — Nanobody de novo Design with AI

Wed 23/09, 13:30 · University of Nova Gorica · Lecture & Hands-on

Nanobodies (Nbs) from pre-immune libraries could exhibit deficiencies in biophysical properties, such as stability and affinity, necessitating an optimization step. This can be achieved either in vitro through controlled mutagenesis followed by another round of panning or in silico by proposing variants. This approach, which involves screening multiple single-site variants, may demand extensive experimental efforts. Additionally, negative epistatic effects can occur when combining multiple potentially beneficial single-site variants, where increased affinity variants may negatively impact stability. Emerging data-driven Machine Learning (ML) approaches are promising in the field of de novo Nb design or redesign, as they can learn multidimensional constraints between protein structure, stability and binding capacity, while explicitly designing protein and non-protein biomolecules at atomic-level resolution. Anyway, a small amount of data is available for training Nbs-specific methods and available models have been tested and validated mainly in the design of mini-binders of known protein biomarkers. In this session an introduction to these methods and their expected accuracy on published data will be provided, with the possibility to try an epitope-targeted nanobody design task through an API-based web service.

### Dr. Claudia d'Ercole — Nanobody Panning Against a Target Antigen

Wed 23/09, 15:30 · University of Nova Gorica · Practical Briefing

In the practical part of the workshop, participants will gain hands-on experience with phage display-based nanobody panning. Working in small groups, they will perform a selection round using a nanobody (VHH) library and a chosen model antigen immobilized on a solid surface. The goal is to enrich for nanobody clones that specifically bind the target. The workflow includes incubation of the library with the antigen, washing to remove non-binders, elution of bound phages, and infection of *E. coli* for amplification. If time allows, participants will also explore panning on live cells to simulate more complex and physiologically relevant selection environments. Takeaway: Participants will understand and carry out the core steps of a nanobody selection campaign, gaining insight into the practical considerations and challenges of identifying specific binders from a large pre-immune or synthetic library.

## Participants talks - Thursday 25/09 · UNG Rožna Dolina campus

### Bor Krajnik — De novo Designed Nanobodies for Intracellular Protein Perturbation

Bor Krajnik<sup>1,2</sup>, Uroš Petrovič<sup>1,2</sup> — <sup>1</sup>Jožef Stefan Institute, Department of Molecular and Biomedical Sciences, Ljubljana, Slovenia; <sup>2</sup>University of Ljubljana, Biotechnical Faculty, Ljubljana, Slovenia

Intracellular binders provide a powerful approach for perturbing protein function. Targeting functional sites, such as active sites and protein–protein interaction interfaces, enables a more nuanced understanding of protein function than gene knockout or knockdown. Nanobodies are attractive scaffolds for such applications as their single-domain architecture and high stability make them well suited for intracellular use. A major limitation in nanobody discovery is the lack of control over the targeted epitope. Advances in deep-learning based structure prediction and protein design have enabled the design of epitope specific nanobodies, addressing this limitation. Using BoltzGen, a generative model for binder design, we have designed nanobodies against a panel of human intracellular protein targets involved in diverse cellular processes, including proteostasis, nuclear transport and translation. The resulting designs will be tested for target binding and functional perturbation as intrabodies in the yeast *Saccharomyces cerevisiae*. By integrating nanobody design with intracellular assays in yeast we aim to expand the number of intrabodies against human proteins.

### Boubekeur Saifi — Generation and Characterization of Disease-Targeting Nanobodies

**Background and Objectives.** This project focuses on the generation and development of single-domain antibody fragments (nanobodies / VHH) as novel tools for the diagnosis and therapy of major human diseases, including cancer and Alzheimer's disease. Due to their small size, high stability, and excellent tissue penetration, these nanobodies offer significant advantages over conventional antibodies for targeted biomedical applications.

**Current Progress & Methodology.** The current phase of the project is dedicated to targeting Galectin-3 (Gal3), a key biomarker associated with disease progression. The project has successfully advanced through the following stages: an immune/synthetic nanobody library was generated (Library Generation & Selection); using the phage display technique, the library was screened against Gal3, leading to the successful selection and isolation of 4 distinct nanobody clusters (Phage Display Screening).

**Current and Future Work.** We are currently in the biophysical characterization phase of these selected nanobodies to evaluate their binding affinity, stability, and specificity toward Gal3. The ultimate goal of this research is to validate these candidates for downstream applications in advanced diagnostics or as therapeutic agents.

### Chaeyeon Park — Engineered DARPIn-Based Imaging Probe Platform for the Detection of Pathological Tau and Tau Variants in Alzheimer's Disease Models

Chaeyeon Park<sup>1</sup>, Rajib Schubert<sup>1,\*</sup> — <sup>1</sup>Department of Bio and Brain Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34141, Republic of Korea

Tau is a soluble microtubule-associated protein essential for maintaining axonal integrity and cytoskeletal stability in healthy neurons. In pathological states such as Alzheimer's disease, hereditary mutations and conformational shifts induce soluble tau to form various pathogenic tau variants and toxic, insoluble filamentous aggregates. Accurate visualization and non-invasive detection of these pathological tau species are crucial for early diagnosis, tracking disease progression, and monitoring therapeutic responses. Although monoclonal antibody-based imaging agents are widely explored, their diagnostic utility is frequently constrained by high molecular weight, poor blood-brain barrier permeability, and prolonged systemic circulation times that lower target-to-background contrast. To overcome these biophysical limitations, we propose a novel molecular imaging strategy utilizing a Designed Ankyrin Repeat Protein (DARPIn)-based probe platform. Characterized by high thermal stability, a small molecular footprint, rapid clearance kinetics, and high modularity, DARPins offer an exceptional scaffold for target-specific molecular imaging. In this study, we systematically engineer the hypervariable binding surfaces of the DARPIn scaffold to generate high-affinity candidates capable of recognizing general tau and distinct pathological tau variants. We evaluate these engineered DARPIn probes in Alzheimer's disease mouse models to assess their brain permeability, binding specificity, and imaging efficacy *in vivo*. This modular platform represents a promising, highly bioavailable diagnostic tool to visualize pathological tau dynamics and advance precision imaging in Alzheimer's disease.

## **Gašper Vrečer — Characterization of Protein–Secondary Metabolite Interactions and Identification of Molecular Binding Determinants**

Secondary metabolites of natural origin represent an important source of bioactive compounds with diverse biological effects, among which antimicrobial activity is particularly prominent. In my PhD project we have so far isolated several groups of secondary metabolites — primarily tannins and carotenoids — from various natural sources, including bacteria and plants. These compounds are systematically investigated for their antimicrobial activity as well as their ability to bind to target proteins.

A central focus of our research is the elucidation of the molecular mechanisms underlying interactions between these isolated compounds and proteins. However, the precise binding sites and structural characteristics of the involved protein motifs remain unknown.

So far, we are analysing the binding of these compounds and model proteins using circular dichroism (CD) spectroscopy, which allows monitoring of changes in the secondary and tertiary structure of proteins upon ligand interaction, and nuclear magnetic resonance (NMR), which allows for determination of binding constants even for weak binders. Using this approach, we will evaluate the binding between proteins and secondary metabolites and assess its impact on protein structure, including thermal stability and denaturation temperature, thereby contributing to a deeper understanding of the molecular mechanisms underlying their biological effects. Based on the preliminary data we can see that the strength of the interactions of these ligands varies based on the surface properties of the model proteins. Next, we want to characterise the structural motifs that are crucial for ligand binding and determine which mutations improve the binding affinity.

To this end we would like to try the use of nanobodies, which represent a powerful tool for the specific recognition and binding of target protein structures, especially since they are small enough to be studied by NMR — in combination with strategies based on affinity chromatography principles, mutated nanobodies can serve as selective ligands for the isolation of proteins with the highest affinity for the studied compounds. This approach enables the enrichment of proteins sharing common structural motifs that are hypothesized to be optimal binding sites for specific secondary metabolites.

## **Mariagrazia Scarano — Functional and Structural Characterisation of Anti-Fyn SH3 VHH and CDR3 Optimization for Improved Intracellular Binding**

Mariagrazia Scarano<sup>1,2</sup>, Elian Dupré<sup>2</sup>, Idir Malki<sup>2</sup>, Justine Mortelet<sup>2</sup>, Céline Reverdy<sup>1</sup>, Isabelle Landrieu<sup>2</sup>, Jean-Christophe Rain<sup>1</sup> — <sup>1</sup>Hybrigenics Services, 1 rue Pierre Fontaine, Evry; <sup>2</sup>Univ. Lille, CNRS UMR9031 – BSI, Integrative Structural Biology, Institut Pasteur de Lille, F-59000 Lille, France

The protein tyrosine kinase (PTK) Fyn is involved in cell proliferation and alternative splicing. Its dysregulation has been linked to carcinogenesis and the generation of abnormally phosphorylated Tau, a hallmark of several neurodegenerative disorders including Alzheimer's disease. The interaction between Tau and the SH3 domain of Fyn has emerged as a potential therapeutic target, as it is implicated in the pathological cascade linking Tau to neuronal damage.

We previously identified H6-1, a VHH against the Fyn SH3 domain, through three rounds of phage display using a synthetic VHH library against full-length mouse Fyn. Epitope mapping by NMR and X-ray crystallography revealed that H6-1 targets the RT loop and N-Src loop of the Fyn SH3 domain, the same region engaged by Tau proline-rich motifs and Sam68. However, H6-1 displayed moderate affinity toward the Fyn SH3 domain, as demonstrated by both yeast surface display and SPR measurements, motivating an affinity maturation campaign focused on CDR3 engineering to identify variants with improved binding properties.

To improve binding properties, a CDR3-focused library was constructed by NNK randomization and screened by yeast surface display over four rounds of FACS-based selection. Among the affinity-matured variants, clone F12 showed a two to 3-fold improvement in binding affinity by SPR, high thermostability, and better intracellular interaction with both Fyn's SH3 and full-length Fyn in yeast two-hybrid assays. NMR competition experiments further demonstrated that F12 prevents the Tau-Fyn SH3 interaction without directly binding Tau, confirming its potential as an intracellular inhibitor of this pathological interaction.

1. Kurochkina, N. & Guha, U. SH3 domains: modules of protein-protein interactions. *Biophys. Rev.* 5, 29-39 (2012).

2. Bielli, P., Busà, R., Paronetto, M. P. & Sette, C. The RNA-binding protein Sam68 is a multifunctional player in human cancer. *Endocr. Relat. Cancer* 18, R91–R102 (2011).

3. Lee, G. et al. Phosphorylation of Tau by Fyn: Implications for Alzheimer's Disease. *J. Neurosci.* 24, 2304–2312 (2004).

### **Perveen Akhtar — Development of a Recombinant Nanobody Pipeline Integrating Computational Design and Phage Display for Sensitive Detection of Microcystin**

Microcystins are toxic cyanobacterial metabolites that can contaminate freshwater systems, aquaculture environments, and seafood, posing significant risks to environmental and food safety. The development of sensitive and specific detection methods requires robust biorecognition elements with high affinity for these toxins. This work aims to develop recombinant nanobodies targeting microcystin through an integrated pipeline combining computational modelling, phage display, recombinant expression, and assay development. Initially, *in silico* approaches, including *de novo* binder design, molecular docking, and molecular simulations, will be used to investigate nanobody–microcystin interactions and identify promising candidate binders. A nanobody phage display library will subsequently undergo biopanning against relevant microcystin antigens, followed by amplification, preliminary ELISA screening, and sequencing of positive clones. The identified nanobody sequences will be further characterized through physicochemical analysis and structural prediction to evaluate properties such as molecular weight, isoelectric point, stability, hydrophobicity, and three-dimensional structure. Promising nanobody candidates will then be subcloned, recombinantly expressed in *Escherichia coli*, purified using affinity chromatography, and characterized by SDS-PAGE. Finally, the selected nanobodies will be evaluated for their applicability in the development of a lateral flow assay for rapid microcystin detection. The developed nanobody-based lateral flow platform could support sensitive and rapid monitoring of microcystin in water, aquaculture environments, and seafood.

### **Taehee Yu — Discovery of TfR1-Binding Fab Fragments**

The blood brain barrier (BBB) is a major obstacle to the delivery of therapeutic biologics to the central nervous system. Receptor-mediated transcytosis, including approaches targeting transferrin receptor 1 (TfR1), offers a potential strategy for facilitating the transport of large therapeutic molecules across the BBB. We are exploring phage display as an approach to identify TfR1-binding Fab fragments that could potentially serve as BBB shuttle modules. TfR1 protein has been prepared as the target antigen, and our planned workflow involves iterative biopanning, phage ELISA, sequence analysis, and recombinant Fab production.

### **Vicky Chouhan — Plant Epigenetics in Conifers: Identification of Epigenetic Biomarkers Under Stress Conditions**

*Vicky Chouhan, PhD Researcher, University of Nova Gorica, Slovenia*

Coniferous trees are long-lived organisms that continuously experience environmental stresses such as drought, temperature fluctuations, salinity, and pathogen pressure. Their ability to survive and adapt to these changing conditions is influenced not only by genetic variation but also by epigenetic mechanisms. Plant epigenetics involves heritable or reversible changes in gene regulation without alterations in the underlying DNA sequence and includes DNA methylation, histone modifications, chromatin remodeling, and regulation by non-coding RNAs. This research aims to investigate stress-associated epigenetic changes in selected conifer species and to identify reliable molecular biomarkers associated with different stress conditions. The study will combine molecular and epigenomic approaches to examine changes in DNA methylation, histone modifications, chromatin structure, and the expression of stress-responsive genes. Comparative analysis of stressed and control plants will be used to identify epigenetic signatures that correlate with specific environmental conditions. The identified biomarkers may provide valuable information about the molecular mechanisms underlying stress adaptation and resilience in long-lived conifers. In the longer term, such biomarkers could contribute to early detection of environmental stress, improved monitoring of forest health, and the development of strategies for conserving and managing conifer populations under changing climatic conditions.

*Keywords: Plant epigenetics; conifers; environmental stress; DNA methylation; histone modification; chromatin remodeling; epigenetic biomarkers; stress adaptation.*

## Invited speakers talks - Friday 25/09 · UNG Rožna Dolina campus

### Assistant Prof. Sara Fortuna — Molecular Dynamics Based Evolution of Epitope Specific Nanobodies

*Fri 25/09, 09:00 · Dipartimento di Scienze Matematiche, Informatiche e Fisiche, Università di Udine · Special Lecture*  
Molecular modelling allows addressing an important bottleneck of antibodies engineering: the control over their target binding site. To reach this goal we have developed an improved parallelised evolutionary algorithm for the computational optimization of generic aminoacid based binders. The code, through a combination of molecular dynamics simulation and free energy estimations, allows to iteratively improve epoch by epoch the predicted binding affinity of a pool of binders. Our result shows the obtained nanobodies capable of capturing the target through the pre-chosen binding site. This holds true even when no prior experimental data is available.

### Assistant Prof. San Hadži, Uroš Zavrtanik, Tadej Medved — Data Quality Challenges and Structural Artifacts in Nanobody–Antigen Databases

*Fri 25/09, 10:00 · Faculty of Chemistry and Chemical Technology, University of Ljubljana · Special Lecture*  
Machine learning methods for protein structure prediction and design depend critically on the quality of the structural data used for training and benchmarking. ML-driven structure prediction of nanobody–antigen complexes is challenging due to the lack of appropriate MSAs to restrain predictions, and large curated structural datasets devoid of structural artifacts could provide data to improve these models. Currently, all nanobody–antigen structural data comes from crystallography, and I will highlight some common artifacts that are present in current structural databases, including incorrectly defined biological assemblies, redundancy bias, crystal contacts mistaken for functional interfaces, packing-induced distortions, and missing density or post-translational modifications near binding sites. I will discuss a new database (AVIDbase) that attempts to address these issues and highlight how much of the nanobody structural space is known and what remains to be discovered.

## Social Dinner & Social Events

### Social Dinner — Šmartno

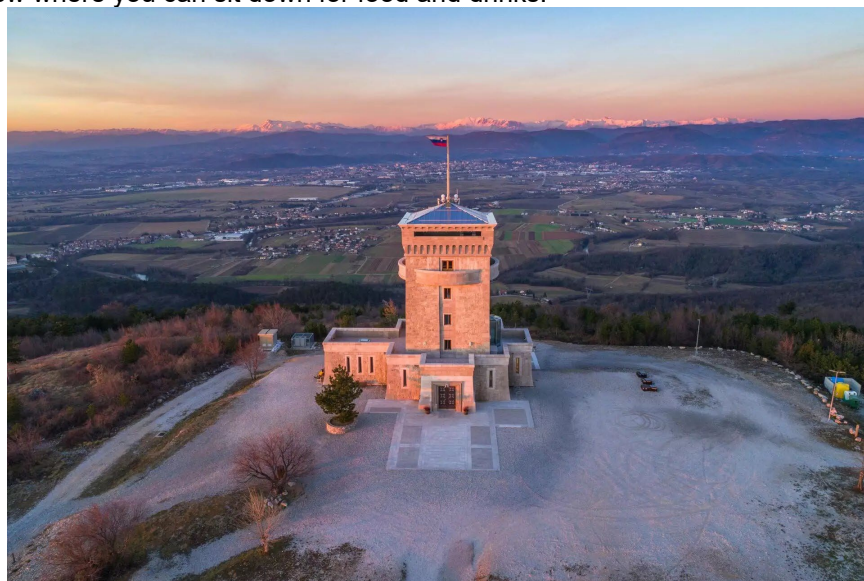
On Thursday evening we'll all head to Šmartno, a picturesque walled medieval village perched on a hilltop in the Goriška Brda wine region. See the schedule for the exact venue and departure details — and [sanmartin.si](http://sanmartin.si) for more on the village and its hospitality.



*Šmartno village at sunset, Goriška Brda. Source: [sanmartin.si](http://sanmartin.si)*

### Friday Trekking — Cerje

On Friday afternoon, once the Summer School wraps up, you may want to join a soft trekking to Cerje, home to the 25-metre Peace Monument — a hilltop tower-museum tracing Slovenian history across seven levels, from the earliest surviving Slovenian texts to the Isonzo Front, with a panoramic terrace over Gorizia, the Karst, the Vipava Valley, the Friuli Plain and the Adriatic Sea. From the UNG Rožna Dolina campus it's about 1h30 on foot to Miren (or ~20 min by bus, line N6237), then a further ~1h, 300 m climb up to Cerje — roughly 2h30 door to door. See the Komoot route for the full trekking details. Right at Cerje there's also a spot with a view where you can sit down for food and drinks.



*Cerje Peace Monument tower at sunset. Source: [go2025.eu](http://go2025.eu)*

### **Gusti di Frontiera**

The Summer School overlaps with "Gusti di Frontiera", the most awaited event of the year for the city of Gorizia: as every year, along the main city streets several food and beverage enogastronomic stands will arrive from all over the world, and will be accompanied with music, fun and shows. An event to not miss!



*Gusti di Frontiera street food festival. Source: [gustidifrontiera.it](http://gustidifrontiera.it)*